

INDIA.RUNS

Build what next India runs on

Team Name : AI Recruiter Intelligence Engine

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Problem Statement: Traditional ATS systems rely on keyword matching, often overlooking qualified candidates. This project builds an AI-powered candidate ranking system that evaluates skills, experience, behavioural signals, and professional relevance to identify the best-fit candidates and generate a trusted shortlist for recruiters.

Solution Overview

- What is your proposed solution?

Develop an AI-powered candidate ranking system that analyzes job requirements and evaluates candidates using skills, experience, behavioral signals, title relevance, and retrieval/search expertise to generate a trusted shortlist of the best-fit candidates.

- What differentiates your approach from traditional candidate matching systems?

Unlike traditional ATS systems that rely on keyword matching, our solution uses a hybrid scoring approach that evaluates the complete candidate profile, including experience, behavioral signals, production expertise, and role relevance, resulting in more accurate and recruiter-friendly candidate recommendations.

JD Understanding & Candidate Evaluation

- What are the key requirements extracted from the JD?
- Strong AI / Machine Learning background
- Experience in Search, Ranking, or Recommendation Systems
- Retrieval and Information Discovery expertise
- Production-level engineering experience
- High relevance between candidate skills and job requirements
- Strong professional and behavioural signals

JD Understanding & Candidate Evaluation

- Which candidate signals are most important for determining relevance? / How does your solution evaluate candidate fit beyond keyword matching

Our solution uses a hybrid scoring approach that evaluates:

- Skill Match Score – Alignment with JD requirements
- Experience Score – Years and level of professional experience
- Behavioural Score – Recruiter engagement and profile signals
- Production Score – Real-world implementation experience
- Title Relevance Score – Alignment of current role with the target role
- Retrieval Relevance Score – Search, ranking, and recommendation expertise

This enables holistic candidate evaluation instead of relying solely on keyword matching.

Ranking Methodology

- How does your system retrieve, score, and rank candidates?
- Parse the Job Description and extract required skills.
- Process candidate profiles and extract relevant features.
- Calculate multiple scoring components:
 - Skill Match Score
 - Experience Score
 - Behavioral Score
 - Production Score
 - Title Relevance Score
 - Retrieval Relevance Score
- Combine the scores using a weighted hybrid ranking formula.
- Sort candidates by final score and generate the Top 100 recommendations.

Ranking Methodology

- What models, algorithms, or heuristics are used?

The solution uses a hybrid rule-based ranking approach with weighted scoring heuristics.

Key techniques include:

- Skill matching against JD requirements
- Experience-based scoring
- Behavioural signal analysis
- Title relevance evaluation
- Retrieval and search expertise detection
- Weighted score aggregation for final ranking

Ranking Methodology

- How are multiple candidate signals combined into a final ranking?

Final Score =

$0.25 \times \text{Skill Score} +$

$0.15 \times \text{Experience Score} +$

$0.15 \times \text{Production Score} +$

$0.15 \times \text{Behavioral Score} +$

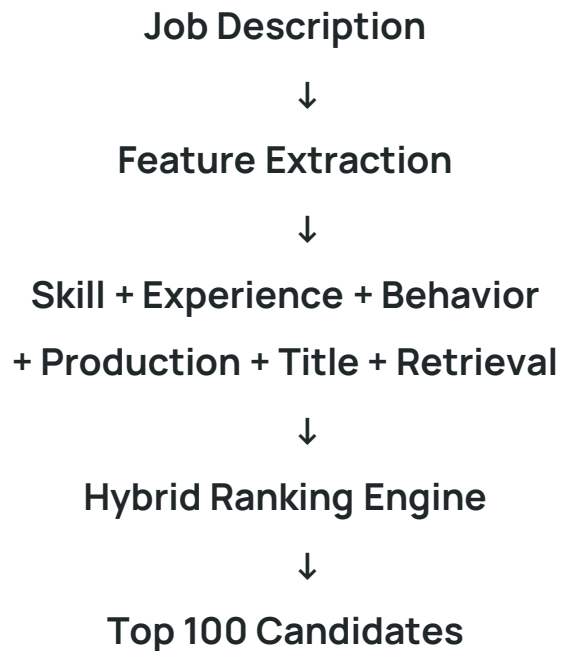
$0.15 \times \text{Retrieval Score} +$

$0.15 \times \text{Title Score}$

Candidates are ranked in descending order of the final score, and the Top 100 candidates are selected for the final shortlist.

Ranking Methodology

Simple Pipeline Diagram:



Explainability & Data Validation

- How are ranking decisions explained?

Each ranked candidate is accompanied by a reasoning summary generated from objective profile signals such as job title, years of experience, recruiter engagement, and role relevance. This provides recruiters with transparent explanations for why a candidate was recommended.

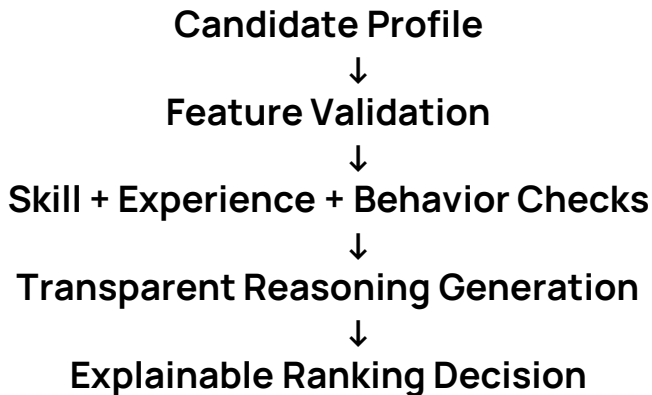
- How do you prevent hallucinations or unsupported justifications?

The system generates explanations only from verified candidate data and computed scoring features. No external information or unsupported assumptions are used, ensuring that all recommendations remain explainable, traceable, and evidence-based.

Explainability & Data Validation

- How does your solution handle inconsistent, low-quality, or suspicious profiles?

The ranking engine evaluates profile quality through multiple signals rather than relying solely on keywords. Profiles with weak experience, low recruiter engagement, poor title relevance, or limited evidence of required skills receive lower scores, reducing the impact of incomplete or potentially misleading profiles.



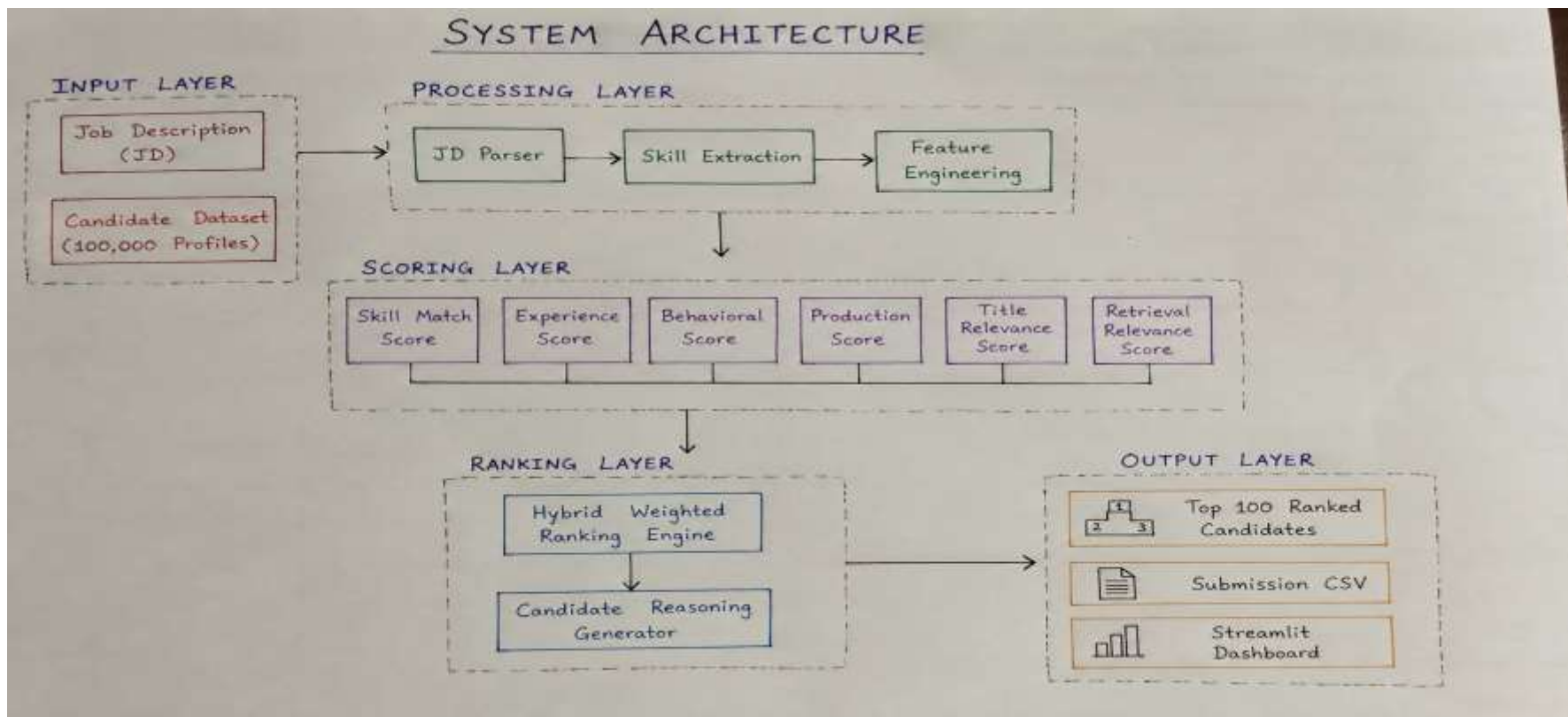
End-to-End Workflow

- What is the complete workflow from JD input to ranked candidate output?

Complete Workflow: JD to Ranked Candidates

1. Job Description (JD) is uploaded and parsed.
2. Required skills and role requirements are extracted.
3. Candidate profiles are loaded from the dataset.
4. Multiple candidate signals are evaluated:
 1. Skill Match Score
 2. Experience Score
 3. Behavioural Score
 4. Production Score
 5. Title Relevance Score
 6. Retrieval Relevance Score
5. A weighted hybrid ranking score is calculated.
6. Explainable reasoning is generated for each candidate.
7. Candidates are ranked by final score.
8. Top 100 candidates are selected and exported as the final submission file

System Architecture



Results & Performance

- What results or insights demonstrate ranking quality?
- Successfully processed 100,000 candidate profiles.
- Generated a ranked shortlist of the Top 100 candidates.
- Prioritised highly relevant profiles such as AI Engineers, Machine Learning Engineers, Search Engineers, NLP Engineers, and Recommendation Systems Engineers.
- Combined multiple candidate signals instead of relying solely on keyword matching.
- Generated transparent reasoning for each recommendation to improve recruiter trust.

Results & Performance

- How does your solution meet the challenge's runtime and compute constraints?
- Runs entirely on CPU without requiring GPUs.
- No external APIs or network calls during candidate ranking.
- Lightweight rule-based scoring ensures efficient processing.
- Handles large-scale candidate datasets while maintaining low memory usage.
- Produces the final ranked output within the challenge's compute constraints.

Technologies Used

- What technologies, frameworks, and tools were used and why were they selected for this solution?

Technology	Purpose
Python	Core Engine
Pandas	Data Processing
Streamlit	Dashboard UI
Matplotlib	Analytics
Python-docx	JD Parsing
NumPy	Score Computation
GitHub	Version Control

Submission Assets

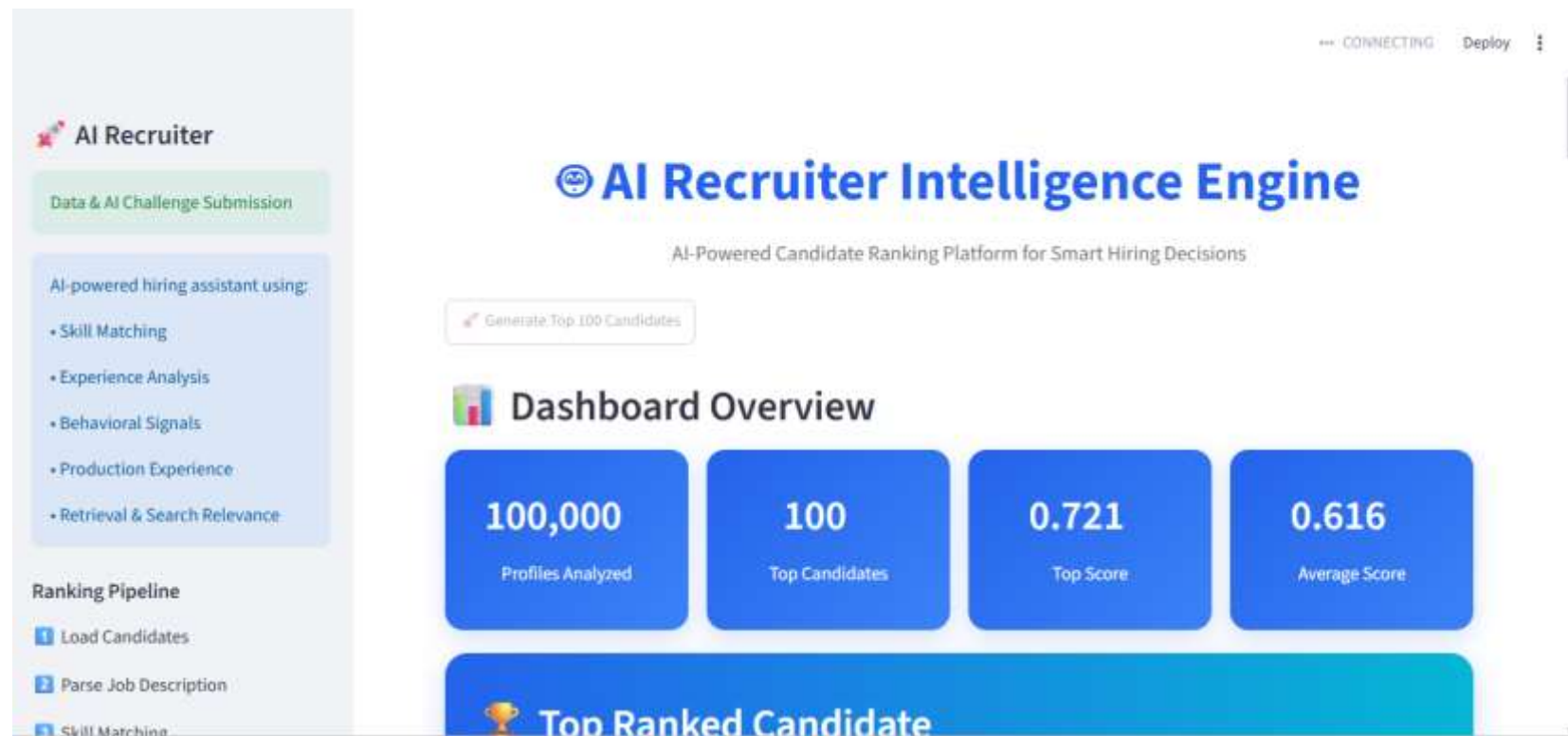
Github Repo Link :- <https://github.com/lokesh2525/AI-Recruiter-Intelligence-Engine.git>

Results & Performance

Dashboard Demonstration

- KPI Cards
- Top Candidate Card
- Analytics Charts
- Top 100 Candidate Table

Results & Performance



Results & Performance

AI Recruiter

Data & AI Challenge Submission

AI-powered hiring assistant using:

- Skill Matching
- Experience Analysis
- Behavioral Signals
- Production Experience
- Retrieval & Search Relevance

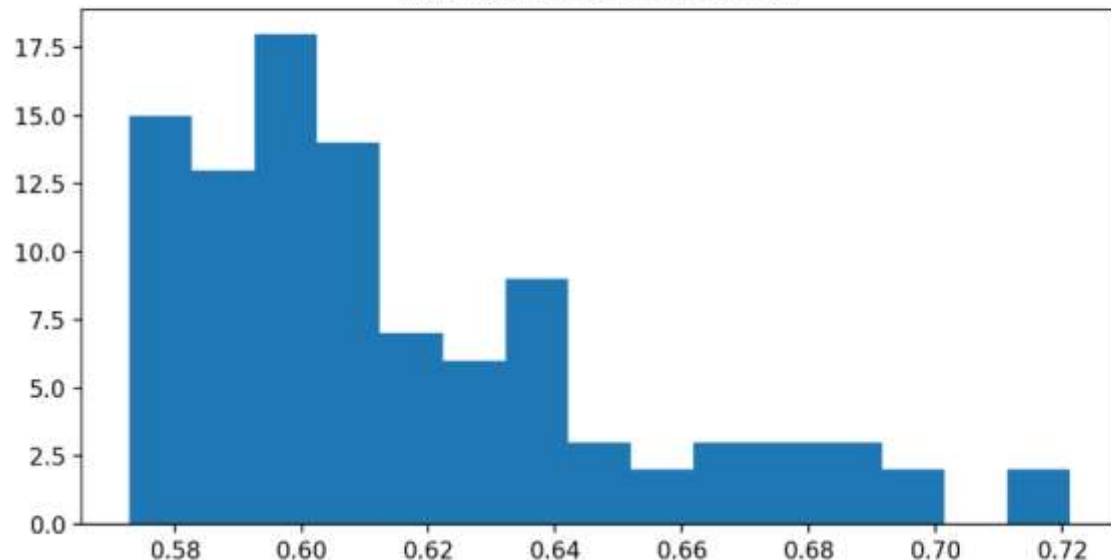
Ranking Pipeline

- 1 Load Candidates
- 2 Parse Job Description
- 3 Skill Matching

CONNECTING Deploy

Score Distribution

Candidate Score Distribution



Results & Performance

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Analytics Candidates Download

Top 100 Ranked Candidates

Search Candidate ID

	Rank	Candidate ID	Title	Experience	Score
0	1	CAND_0041669	Recommendation Systems Engineer	8	0.7212
1	2	CAND_0071974	Senior AI Engineer	7.8	0.717
2	3	CAND_0077337	Staff Machine Learning Engineer	7	0.6987
3	4	CAND_0002025	Senior AI Engineer	5.9	0.6974
4	5	CAND_0011162	Recommendation Systems Engineer	5.8	0.6908
5	6	CAND_0077285	Recommendation Systems Engineer	5.5	0.69
6	7	CAND_0026532	Recommendation Systems Engineer	4.8	0.6859
7	8	CAND_0018499	Senior Machine Learning Engineer	7.2	0.6785
8	9	CAND_0042506	Search Engineer	4.2	0.6785
9	10	CAND_0001206	Search Engineer	4	0.6785

Results & Performance

 AI Recruiter

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- Skill Matching
- Experience Scoring
- Behavioral Signals
- Production Experience
- Retrieval/Search Relevance
- Title Relevance

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THANK YOU